

Q1: 24 Feb (Shift 2) - Numerical

The solubility product of PbI_2 is 8.0×10^{-9} . The solubility of lead iodide in 0.1 molar solution of lead nitrate is $x \times 10^{-6}$ mol/L. The value of x is _____ (Rounded off to the nearest integer) [Given $\sqrt{2} = 1.41$]

Q2: 25 Feb (Shift 1) - Single Correct

The solubility of AgCN in a buffer solution of $\text{pH} = 3$ is x. The value of X is:

[Assume: No cyano complex is formed; $K_{\text{sp}}(\text{AgCN}) = 2.2 \times 10^{-16}$ and $K_{\text{a}}(\text{HCN}) = 6.2 \times 10^{-10}$]

(1) 0.625×10^{-6}

(2) 1.6×10^{-6}

(3) 2.2×10^{-16}

(4) 1.9×10^{-5}

Q3: 25 Feb (Shift 2) - Single Correct

The solubility of $\text{Ca}(\text{OH})_2$ in water is :

[Given : The solubility product of $\text{Ca}(\text{OH})_2$ in water = 5.5×10^{-6}]

(1) 1.11×10^{-6}

(2) 1.77×10^{-6}

(3) 1.77×10^{-2}

(4) 1.11×10^{-2}

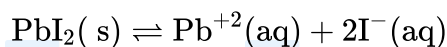
Q4: 26 Feb (Shift 2) - Numerical

The pH of ammonium phosphate solution, if pK_{a} of phosphoric acid and pK_{b} of ammonium hydroxide are 5.23 and 4.75 respectively, is

Answer Key**Q1 (141)****Q2 (4)****Q3 (4)****Q4 (7)**

Q1 (141)

$$K_{SP}(\text{PbI}_2) = 8 \times 10^{-9}$$



$$S + 0.1 \quad 2S$$

$$K_{SP} = [\text{Pb}^{2+}] [\text{I}^{-}]^2$$

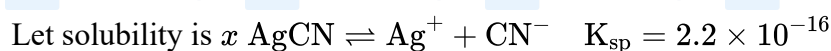
$$8 \times 10^{-9} = (S + 0.1)(2S)^2 \Rightarrow 8 \times 10^{-9} \simeq 0.1 \times 4S^2$$

$$\Rightarrow S^2 = 2 \times 10^{-8}$$

$$S = 1.414 \times 10^{-4} \text{ mol/Lit}$$

$$= x \times 10^{-6} \text{ mol/Lit} \quad \therefore x = 141.4 \simeq 141$$

Q2 (4)



$$K_{sp} \times \frac{1}{K_a} = [\text{Ag}^{+}] [\text{CN}^{-}] \times \frac{[\text{HCN}]}{[\text{H}^{+}][\text{CN}^{-}]}$$

$$2.2 \times 10^{-16} \times \frac{1}{6.2 \times 10^{-10}} = \frac{[S][S]}{10^{-3}}$$

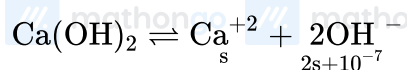
$$S^2 = \frac{2.2}{6.2} \times 10^{-9}$$

$$S^2 = 3.55 \times 10^{-10}$$

$$S = \sqrt{3.55 \times 10^{-10}}$$

$$S = 1.88 \times 10^{-5} \Rightarrow 1.9 \times 10^{-5}$$

Q3 (4)



$$s(2s + 10^{-7})^2 = 55 \times 10^{-7}$$

$$4s^3 = 55 \times 10^{-7}$$

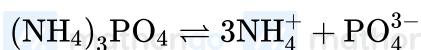
$$s^3 = \frac{5500}{4} \times 10^{-9}$$

$$s = \left(\frac{2250}{2}\right)^{1/3} \times 10^{-3}$$

$$s = (1125)^{1/3} \times 10^{-3}$$

$$s = 1.11 \times 10^{-2}$$

Q4 (7)



$$[\text{H}^+] = K_a \times \sqrt{\frac{K_w}{K_a \times K_b}}$$

$$\text{pH} = \text{p}K_a + \frac{1}{2} \{ \text{p}K_w - \text{p}K_a - \text{p}K_b \}$$

$$\text{pH} = 5.23 + \frac{1}{2} \{ 14 - 5.23 - 4.75 \}$$

$$\text{pH} = 5.23 + \frac{1}{2} (4.02) = 7.24 = 7 \text{ (Nearest integer)}$$